

Towards the Development of Transparent Graphene Micro Electrode Arrays

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ABSTRACT

Microelectrode arrays (MEAs) [1] are widely used to record the cellular activity. The standard electrode materials in use are platinum, gold and titanium nitride. The entry of carbon based materials in the field of MEAs is only a decade old. Graphene and carbon nanotubes (CNTs) are extensively researched carbon based materials. In our previous work semi-transparent CNT MEAs are reported to give excellent impedance [2]. Application of graphene in optogenetics [3] and calcium imaging [4] is reported in the year 2014. The main advantage of graphene is transparency, which comes at the cost of impedance. Here, we present the fabrication of graphene-based microelectrode arrays comprising gold interconnects terminated by graphene electrodes, where an optimal balance between transparency, size of the electrode and impedance is achieved.

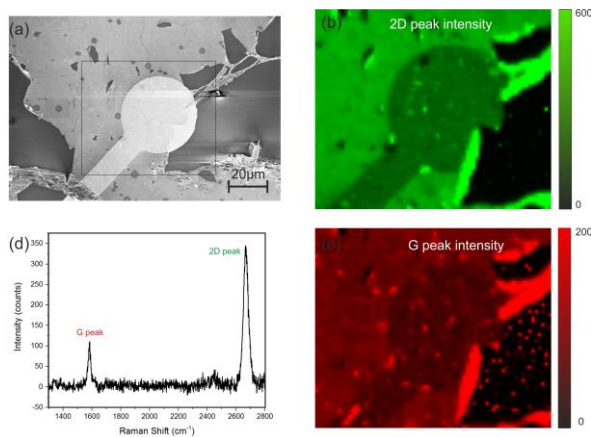


Fig 1: (a) Scanning electron micrograph of monolayer graphene on Au micro electrode.

(b) 2D peak intensity map over the region indicated in (a).

(c) G peak intensity map over the region indicated in (a).

(d) Raman spectrum confirming the presence of monolayer graphene.

References

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